



Mohammad Rouie Miab

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Summary

Machine learning and research engineer with experience in computer vision, generative models, multimodal learning, attention mechanisms, and vision-language alignment. M.S. in Computer Science with thesis research on inference-time latent optimization for semantic precision in diffusion models.

Education

Washington University in St Louis

Jan 2025 – May 2026

Computer Science, M.S. – Thesis-based

Topic: *Improving Semantic Precision in Diffusion Models via Latent Optimization*

Missouri University of Science and Technology

Aug 2019 – Dec 2023

Computer Science, B.S., Computer Engineering, B.S.

Research

Improving Semantic Precision in Text-to-Image Diffusion Models

Aug 2025 – May 2026

Master's Thesis, Washington University in St. Louis

- Reviewed existing literature on semantic precision and diffusion latent optimization
- Collaboratively designed and implemented a training-free inference-time latent optimization pipeline to mitigate attribute leakage and object overlap in diffusion models
- Built upon Attend-and-Excite's (AE) work with our noun-phrase-level binding and separation losses on cross-attention maps, using JSD-based spatial separation and centroid repulsion
- Conducted qualitative and benchmark evaluations on hard prompts, extending prior work
- Demonstrated significantly improved compositional robustness compared AE and baselines
- Identified failure modes in CLIP-based metrics' compositional and semantic correctness

Experience

The Boeing Company — St. Louis, MO

Jun 2023 – Aug 2024

Government Training Engineering Software Intern, Agile Tools Team (extended)

- Developed utilities for seamless large-scale project data manipulation and migration
- Crafted and maintained code for advanced project analytics and automation

Missouri S&T Multirotor Design Team — Rolla, MO

Jan 2020 – Dec 2023

Chief Software Engineer, CV Engineer, Simulation Lead, and other roles

- Led software team, spearheading and roadmapping design and development with agile methods, while facilitating team training, task allocation, and project delivery
- Innovated object detection processes, improving reliability against complex backgrounds
- Developed an obstacle persistence algorithm, providing vision avoidance capabilities
- Optimized text detection algorithms, improving accuracy in challenging conditions
- Led the Simulation sub-team, using Unreal Engine 4 & AirSim to enable virtualized flight tests

Skills

Languages and Tools:

Python, Torch, TensorFlow, C/C++/C#, JS, Rust

Technical Skills:

Computer vision, diffusion models, attention mechanisms, multimodal language models, LSTMs

Projects

- **PET-MRI Parkinson's** – Classifying PD severity using F-FDG-PET and synthetic-generated fMRIs
- **Thermo-Fisher Scientific Robotic Arm** – Auto object localization & tracking for assembly lines
- **AI Stock Trading** – Custom LSTM model for predictive trading based on historical time-series data
- **Multirotor UAV Vision Projects** – Object & obstacle classification, tracking, and avoidance systems